5 Reflection and transmission at an interface

5.1 Introduction

The smiling baby in Figure 5.1 was not yet born at the time his picture was taken. The method used to produce the image relies on an interesting property of sound waves: When a wave travelling in a certain medium reaches a discontinuity in the properties of the medium, it splits into a reflected wave and a transmitted wave. In this case, an ultrasound wave is reflected from the baby's face because there is an abrupt change in the acoustic properties of the medium there. The image is built up from such reflected waves. In this chapter, we will explore why this happens, and what are the properties of the reflected and transmitted waves. We will work with the case of sinusoidal waves, but the results may be extended to all types of waves, including pulses.



Figure 5.1: Ultrasound fetal image – Terry M tmat1075

5.2 Reflection and transmission coefficients

Figure 5.2 shows the situation schematically. The dashed line represents the interface between mediums 1 and 2. Three different waves appear at the interface, which for mathematical convenience we represent as complex functions. An **incident wave** $\tilde{y}_i(x,t)$ travels in the $+x$ direction. At the interface, it generates a **reflected wave** $\tilde{y}_r(x,t)$ travelling in the $-x$ direction in the original medium, and a **transmitted wave** $\tilde{y}_t(x,t)$ travelling in the $+x$ direction in medium 2.

The incident wave may be expressed as

$$\tilde{y}_i = A_i e^{i(\omega t - k_1 x)} \tag{5.1}$$

We choose $x = 0$ at the interface. We also choose $t = 0$ so that the initial phase is zero, and the amplitude A_i is therefore real. Notice the minus sign in the phase, which indicates that the wave is travelling in the $+x$ direction. Also, the wave number has a subscript 1 because it depends on the medium, but the frequency has no subscript because it does not change from one medium to the other.

Similarly, we can write expressions for the reflected and transmitted

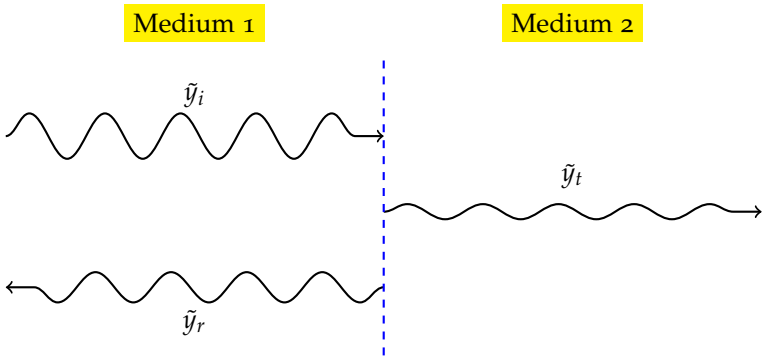


Figure 5.2: A diagram of the waves arising at an interface between two different media.

waves:

$$\tilde{y}_r = \tilde{A}_r e^{i(\omega t + k_1 x)} \tag{5.2a}$$

$$\tilde{y}_t = \tilde{A}_t e^{i(\omega t - k_2 x)} \tag{5.2b}$$

The amplitudes of the reflected and transmitted waves may be complex, as they may differ in phase from each other and from the incident wave.

The question that we address here is: What are the reflected and transmitted amplitudes, in relation to the incident amplitude? For this, we also need some information about what is going on at the interface. There are two continuity conditions that the waves must meet:

Condition 1: The value of $\tilde{y}$ on the left hand side of the interface must be the same as that on the right hand side. If we think of two strings with different densities joined together at $x = 0$, for example, this condition is equivalent to saying that there must not be a gap in the strings there. This condition may be written as

$$\tilde{y}_i + \tilde{y}_r = \tilde{y}_t \text{ at } x = 0 \tag{5.3}$$

Substituting the expressions for the $\tilde{y}$'s and simplifying, this is equivalent to

$$A_i + \tilde{A}_r = \tilde{A}_t \tag{5.4}$$

Condition 2: The first derivative of $\tilde{y}$ on the left hand side of the interface must be the same as that on the right hand side. Again, if we think of two strings joined together, this would mean that there is no kink in the string at $x = 0$. If there was, there would be a net shearing force on the string there, which is not possible. So we have that

$$\tilde{y}_i' + \tilde{y}_r' = \tilde{y}_t' \text{ at } x = 0 \tag{5.5}$$

and once again substituting the expressions for the $\tilde{y}$'s and simplifying, this is finally expressed as

$$-k_1 A_i + k_1 \tilde{A}_r = -k_2 \tilde{A}_t \quad \text{at } x = 0 \tag{5.6}$$

Now we define the **reflection** and **transmission coefficients**:

$$\mathcal{R} = \frac{\tilde{A}_r}{A_i} \tag{5.7a}$$

$$\mathcal{T} = \frac{\tilde{A}_t}{A_i} \tag{5.7b}$$

Then we can rewrite the two conditions (5.4) and (5.6) in terms of $\mathcal{R}$ and $\mathcal{T}$:

$$1 + \mathcal{R} = \mathcal{T} \tag{5.8a}$$

$$1 - \mathcal{R} = \frac{k_2}{k_1}\mathcal{T} \tag{5.8b}$$

which we can solve for the coefficients:

$$\mathcal{T} = \frac{2}{1 + \frac{k_2}{k_1}} \tag{5.9}$$

and

$$\mathcal{R} = \frac{1 - \frac{k_2}{k_1}}{1 + \frac{k_2}{k_1}} \tag{5.10}$$

Finally, it is more instructive to express the coefficients in terms of the impedance Z rather than the wave number k. In general, Z is inversely proportional to the wave speed v[1], and $v = \omega/k$, so Z is directly proportional to k. Therefore we can substitute Z_2/Z_1 for k_2/k_1, to arrive at

[1] See equations (4.19) and (4.28).

$$\mathcal{T} = \frac{2Z_1}{Z_1 + Z_2} \tag{5.11a}$$

$$\mathcal{R} = \frac{Z_1 - Z_2}{Z_1 + Z_2} \tag{5.11b}$$

To better understand what these coefficients are telling us about the reflected and transmitted waves, let's look at some particular cases.

$Z_1 = Z_2$: When the impedances are equal, $\mathcal{R} = 0$, and $\mathcal{T} = 1$. In other words, there is no reflected wave, and the incident wave is fully transmitted. Notice that this doesn't mean that the two media are the same, only that they have the same impedance.

A good example of this is the instrument used to make the ultrasound image of Figure 5.1. The end of an ultrasound probe that is in

contact with the skin (see Figure 5.3) is made of a material that has the same acoustic impedance as the tissue, and a gel is applied to the surface that ensures good contact, avoiding jumps in impedance. The result is that there are no reflections at the point of contact, so that the tissue is effectively transparent to the sound waves.



Figure 5.3: An ultrasound probe.

$Z_2 \gg Z_1$: As we saw in the previous chapter, a large value of Z means that the medium offers great resistance to the passage of a wave. To take this to an extreme, imagine a string tied to a wall. In comparison to the string, the wall has a practically infinite impedance. In this case, we would expect all of an incident wave to be reflected, and none of it to be transmitted into the wall, or a sound wave hitting a wall. And indeed, in the limit $Z_2/Z_1 \to \infty$, the transmission coefficient $\mathcal{T}$ can be seen to tend to zero.

More interesting is that the reflection coefficient $\mathcal{R}$ tends to -1. Recall that $\mathcal{R} = \tilde{A}_r/A_i$. $\tilde{A}_r$ is the *complex* amplitude of the reflected wave, that is

$$\tilde{A}_r = A_r e^{i\phi_r} \tag{5.12}$$

where ϕ_r is the initial phase of the reflected wave, relative to the initial phase of the incident wave, which we chose to be zero. So ϕ_r is the phase difference between the reflected wave and the incident wave. The reflection coefficient can also be written as

$$\mathcal{R} = \frac{|\tilde{A}_r|}{|\tilde{A}_i|} e^{i\phi_r} \tag{5.13}$$

implying that $e^{i\phi_r} = -1$ and therefore $\phi_r = \pi$. So as we expected, the ratio of the *magnitudes* of the amplitudes is 1, meaning that the amplitude of the reflected wave is the same as that of the incident wave. But the surprise is that the reflected wave is phase shifted by π relative to the incident wave. We will explore an important instance of this in the next section.

$Z_2 \ll Z_1$: This the opposite case to the previous one. Imagine now shaking a rope with a free end, or the sound wave emerging from the open end of a wind instrument: The impedance of the second medium is negligible compared with the first. If we let Z_2/Z_1 tend to zero, the reflection coefficient becomes 1, and the transmission coefficient becomes 2. This is a very counterintuitive result: Apparently, there is a reflected wave at the opening that is identical to the incident wave, and there is a transmitted wave that has twice the amplitude of the incident wave!

To make sense of this result, we need to explore not just the amplitudes of the reflected and transmitted waves, but how much power they transport. We turn to the matter of energy next.

It is worth noting that in the case when $Z_2 \ll Z_1$, the reflection coefficient is positive, indicating that the reflected wave is in phase with the incident wave. This in contrast to the case where $Z_2 \gg Z_1$.

5.3 *The energy balance at the interface between two media*

In general, the power transported past a point by a wave of amplitude A and frequency ω in a medium with impedance Z is

$$P = \frac{1}{2} Z \omega^2 A^2 \tag{5.14}$$

Let's define reflection and transmission coefficients for energy. The reflection coefficient $\mathcal{R}_e$ is the ratio between the power passing through the boundary at $x = 0$ due to the reflected wave, and the power due to the incident wave:

$$\mathcal{R}_e = \frac{\frac{1}{2} Z_1 \omega^2 |\tilde{A}_r|^2}{\frac{1}{2} Z_1 \omega^2 A_i^2} = \frac{|\tilde{A}_r|^2}{A_i^2} = \mathcal{R}^2 \tag{5.15}$$

Similarly, the energy transmission coefficient is

$$\mathcal{T}_e = \frac{\frac{1}{2} Z_2 \omega^2 |\tilde{A}_t|^2}{\frac{1}{2} Z_1 \omega^2 A_i^2} = \frac{Z_2 |\tilde{A}_t|^2}{Z_1 A_i^2} = \frac{Z_2}{Z_1} \mathcal{T}^2 \tag{5.16}$$

So in the case that $Z_2 \ll Z_1$, where we found $\mathcal{R} = 1$ and $\mathcal{T} = 2$, we find

$$\mathcal{R}_e \approx 1$$
$$\mathcal{T}_e \approx 0$$

The reflected wave carries most of the energy of the incident wave, and the transmitted wave carries practically no energy.

Example 5.1: *A source of plane waves emits* $1.00 \times 10^{-6}\,\mathrm{W\,m^{-2}}$ *of power per square metre. The waves travel a distance d in a medium with acoustic impedance* Z_1*, until they reach another medium with acoustic impedance* $Z_2 = 3Z_1$*. What are the transmitted and reflected intensities at the boundary?*

For plane waves, the intensity does not depend on distance from the source, so the incident intensity at the boundary is also $1.00 \times 10^{-6}\,\mathrm{W\,m^{-2}}$. The energy reflection coefficient is

$$\mathcal{R}_e = \mathcal{R}^2 = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2 = \frac{1}{4}$$

so the reflected intensity is

$$I_r = 2.50 \times 10^{-7}\,\mathrm{W\,m^{-2}}$$

and therefore the rest is transmitted:

$$I_t = 7.50 \times 10^{-7}\,\mathrm{W\,m^{-2}}$$

5.4 *Standing waves*

Let us return to the case where $Z_2/Z_1 \to \infty$. If we think of a vibrating string, in this case one end of the string would be fixed to a wall. Or, if we think of a plane sound wave, it would be incident on a very massive wall.

In medium 1, where the incident wave comes from, the reflected wave is also travelling in the opposite direction. So in fact in medium 1, we have a superposition of these two waves. To calculate the resulting wave, we resort to the usual technique of assuming that y_i is a cosine function and think of it as the real part of a complex function; then we calculate the superposition of the incident and reflected waves as complex functions, because it's simpler mathematically, and finally take the real part of the result. So we start with an incident wave

$$\tilde{y}_i = A_i e^{i(k_1 x - \omega t)} \tag{5.18}$$

We saw in the previous section that in this case the reflection coefficient is -1, which tells us that $\tilde{A}_r = -A_i$. So we can write the reflected wave as

$$\tilde{y}_r = -A_i e^{i(k_1 x + \omega t)} \tag{5.19}$$

We can call the superposition simply $\tilde{y}_1$ because it is the complete wave that is actually propagating in medium 1. Adding the incident and reflected waves, this turns out to be

$$\begin{aligned} \tilde{y}_1 = \tilde{y}_i + \tilde{y}_r &= A_i e^{ikx}\left(e^{-i\omega t} - e^{i\omega t}\right) \\ &= A_i e^{ikx}(-2i\sin\omega t) \end{aligned} \tag{5.20}$$

Now we can take the real part to get the real wave in medium 1:

$$y_1 = 2A_i \sin\omega t \sin kx \tag{5.21}$$

This has an interesting interpretation: If we take successive snapshots of the wave, they are all of the form

$$A(t)\sin kx \tag{5.22}$$

which is a sine function in space with an amplitude that fluctuates in time $A(t) = 2A_i \sin\omega t$. Because $\sin kx$ does not depend on time,

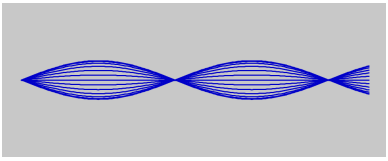


Figure 5.4: Snapshots of a standing wave caused by reflection at a fixed end

the effect is that of a wave that is standing still, with a fluctuating amplitude. This is called a **standing wave**.

Figure 5.4 shows a series of periodic snapshots of the standing wave, with a fluctuating amplitude. At certain points, y is zero at all times. According to equation (5.22), those are the points where $\sin kx = 0$. Those particular points along the string are called **nodes**, and the points in between where y is maximum are called **antinodes**.

At the opposite extreme where $Z_2/Z_1 \to 0$ (a string with a free end, or a sound wave emerging from a tube with an open end), the result is also a standing wave. This time the reflection coefficient is $\mathcal{R} = 1$, so

$$\tilde{y}_r = A_i e^{i(kx+\omega t)} \tag{5.23}$$

and the superposition is

$$y_1 = 2A_i \cos \omega t \cos kx \tag{5.24}$$

Again, we have a stationary wave form in space with an amplitude that fluctuates in time – a standing wave.